Incident Investigation Report: PI-0001

AI-Assisted Video Risk Assessment & Evidence Extraction Prototype

PROTOTYPE REPORT NOTICE: Measurements reflect uncalibrated 2D image coordinates. Proximity and closing vectors indicate candidate interactions for review, not certified physical collisions.

1. Incident Summary & Timeline

The automated trajectory analysis identified an approaching interaction between tracked road entities. This event record serves as decision support for human investigators, not as an autonomous collision certainty or legal determination.

Active Episode Interval: 5.47s to 5.50s | **Duration:** 0.03 seconds | **Closest-Approach Timestamp:** 5.500 seconds

2. Kinematic & Observational Metrics

Investigation Attribute	Recorded Algorithmic Metric
Incident Dossier ID	PI-0001
Target Timestamp	5.500 seconds
Classification Finding	POTENTIAL_APPROACHING_INTERACTION
Algorithmic Confidence	UNVALIDATED_HEURISTIC
Tracked Entity IDs	168, 185
Consecutive Detections	3
Peak Consecutive Confirmations	5
Minimum Image-Space Distance	27.78 px
Peak Image-Space Closing Speed	247.72 px/s
Operator Triage Decision	PENDING REVIEW
Authority Dispatch Status	NOT SUBMITTED

3. Explainable Risk Factor Breakdown

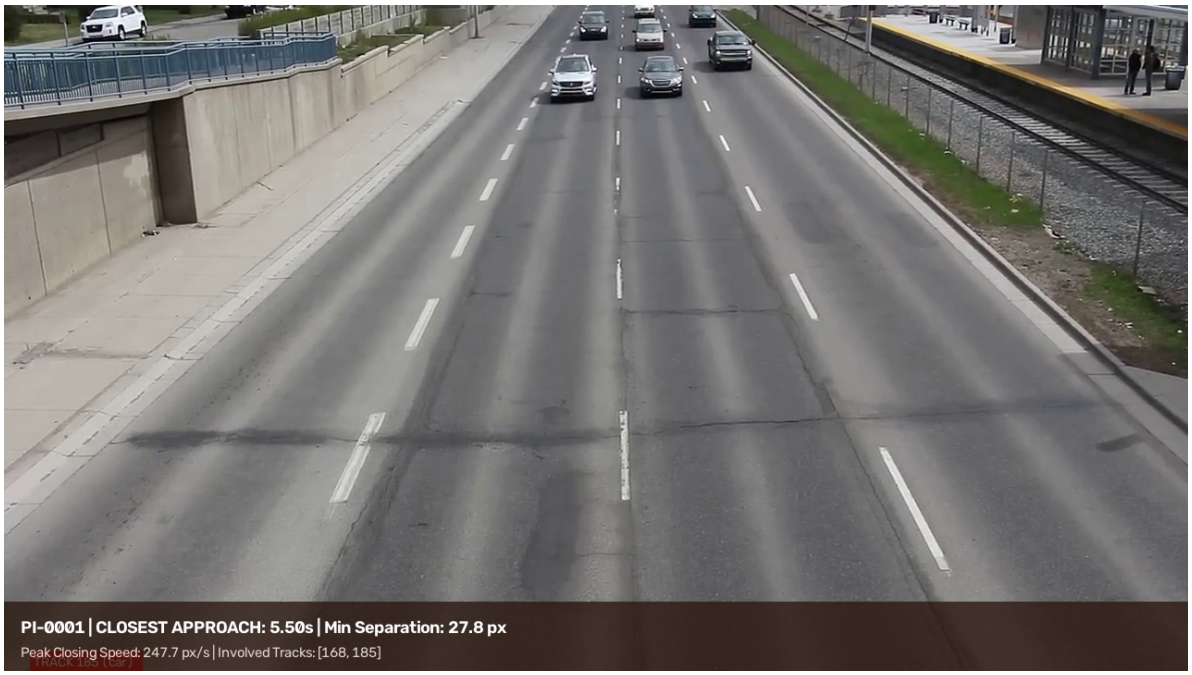
Algorithmic decision-support factors explaining how the system characterized event risk:

Evaluation Factor	Algorithmic Finding	Investigation Rationale & Sensor Context
Potential path conflict	**Detected**	Spatial-temporal trajectory projection indicates intersecting vehicle paths within review window.
Close proximity	**Detected**	Severe spatial proximity: closest center distance reached 27.8 px (< 15 px threshold).
Rapid approach	**Detected**	Rapid approach confirmed: peak closing speed measured at 247.7 px/s.

Evaluation Factor	Algorithmic Finding	Investigation Rationale & Sensor Context
Collision confirmation	**Unavailable**	Candidate anomaly flagged; monocular 2D overlap does not prove physical contact or vehicle deformation; human triage required.

4. Supporting Visual Evidence

Frame captured at closest approach (5.500 seconds) with tracked entity bounding boxes and IDs.



5. Technical Limitations & Required Next Steps

• **Limitation:** Image-space measurements are not real-world distances.

• **Limitation:** A potential interaction does not establish a collision.

• **Action:** Cross-reference timestamp with secondary camera viewpoints or roadside telemetry.

• **Action:** Perform optical camera calibration if real-world velocity measurements are required for litigation.

• **Action:** Submit incident dossier to designated authority dispatcher for manual verification.